

The empirical recurrence for OEIS A237213: recovering a conjecture that is not written down

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Abstract

OEIS A237213 records “Empirical recurrence of order 58”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 256$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 58, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 58 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A237213 is “Number of $(n+1) \times (2+1)$ 0..3 arrays with the maximum plus the upper median minus the lower median minus the minimum of every 2×2 subblock equal.”. It has offset 1 and begins

1076, 6200, 37858, 246916, 1641592, 11208306, 77496396, 543475684, ...

The entry states:

Empirical recurrence of order 58 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 09:58:15, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 256$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 256$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 512$ exact terms of the model returns order 58 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{58} c_i a(n-i),$$

c_1	37	c_{16}	−67238651730680	c_{31}	−31697782876596016672	c_{46}	6204168108608307200
c_2	−395	c_{17}	54665757873771	c_{32}	68229070158574610952	c_{47}	2613981534591854182
c_3	−1800	c_{18}	772129870862862	c_{33}	43422841038778969624	c_{48}	−618629691603838566
c_4	65243	c_{19}	−1477077800189721	c_{34}	−160012552950958064232	c_{49}	−484490367618514944
c_5	−282441	c_{20}	−6154763850166932	c_{35}	−15905875686374846416	c_{50}	1781603031480532992
c_6	−3208488	c_{21}	19082608257626247	c_{36}	266196374066908741472	c_{51}	483478244529537024
c_7	31523347	c_{22}	32088045228888569	c_{37}	−71740791713958691584	c_{52}	−24784262125741670
c_8	34751020	c_{23}	−163286931484967928	c_{38}	−311820118911388602944	c_{53}	−23837294658060288
c_9	−1415882425	c_{24}	−78843967614647559	c_{39}	177740615096055372928	c_{54}	18192331085709312
c_{10}	2871106361	c_{25}	996778195731449961	c_{40}	248688657598617686016	c_{55}	456162973581312
c_{11}	35563572140	c_{26}	−274680325641776208	c_{41}	−219235653087040427008	c_{56}	−683478010036224
c_{12}	−155046174246	c_{27}	−4428293527268866207	c_{42}	−123744746997180733440	c_{57}	0
c_{13}	−506729305984	c_{28}	3811230736171170891	c_{43}	168663429593488343040	c_{58}	10436770529280
c_{14}	4032646846699	c_{29}	14212858616957451370	c_{44}	27625825509798404096		
c_{15}	2506477275369	c_{30}	−20216007863027269248	c_{45}	−83637509731008430080		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 58, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $58 + 4$ terms removed returns order 58 as well, so $\deg q_{\min} = 58 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 18 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $58 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 58 that this sequence satisfies — and that family turns out to have exactly one member.

References

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